

Water quality testing

Testing Document

Version 1

Date:

- Start: 01.07.2025
- End: 25.09.2025

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General description and purpose

Purpose of test

The sink-float separation is likely one of the methods used for the separation of plastic types in Blantyre. In the case that the results acquired during small scale tests in Zurich are not reproducible during the field work in Malawi, possible influence factors must be examined. For this, the water is analyzed, first in Zurich, then at Waste Adviser's Innovation Hub in Blantyre.

Measured values of the water are:

- pH value
- (Total) water hardness
- Free chlorine content
- Iron content
- Copper content
- Lead content
- Nitrate content
- Nitrite content
- Potassium monopersulfate content
- Total chlorine content
- Fluoride content
- Cyanuric acid content
- Ammonium chloride content
- Bromines content
- Total alkalinity
- Carbonates content

These values can serve for comparison during the analysis.

Description of Test

For the pH testing, the [LLG-pH Meter 5](#) is used. Furthermore, a [water hardness test set](#) and a [16 in 1 water quality tester](#) is acquired. For all measurements, tap water from the same source as for the tests is used. The test procedures follow the instructions included with each product.

Calculations

Table 1: Water hardness unit conversion ("Wasserhärte messen," n.d.)

mol/m ³		<0.7	>1.0	>1.8	>2.7	>3.6	>4.5
°d	German degree of hardness	0	5	10	15	20	25
°f	French degree of hardness	0	9	18	27	36	45
°e	English degree of hardness	0	6.25	12.5	18.75	25	31.25

Schedule of test with all steps required

1. Read the instructions for all three test devices.
2. To obtain the test water let the water run until a constant temperature is reached. Then, fill two beakers for the tests.
3. Conduct the pH measurement with the pH meter by following the instructions of the device.
4. Set up the camera for the hardness test.
5. Conduct the water hardness measurement by following the tools instructions.
6. Conduct the water quality test by following its instructions.
7. Record a photo of the result.
8. Repeat the steps for all water sources used for testing.

Risk analysis

Potential risks associated with this test

None.

Mitigation steps

Safety glasses are worn during the whole process.

Risks after taking mitigation steps

None.

Tools and devices required

Tool	Check-in tick
1x safety glasses	X
Paper towels	X
1x laptop	X
1x camera (phone)	X
1x timer (laptop or phone can be used)	X
1x pH meter	X
1x water hardness test set	X
1x 16 in 1 water quality tester	X
2x beaker	X
0.8 l tap water	X

Testing Results

All recordings and pictures of the process and the results for each sample are available [here](#).

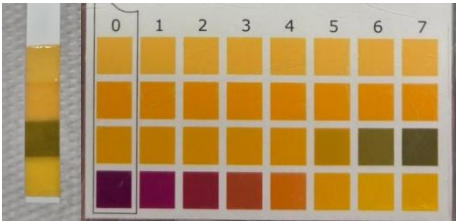
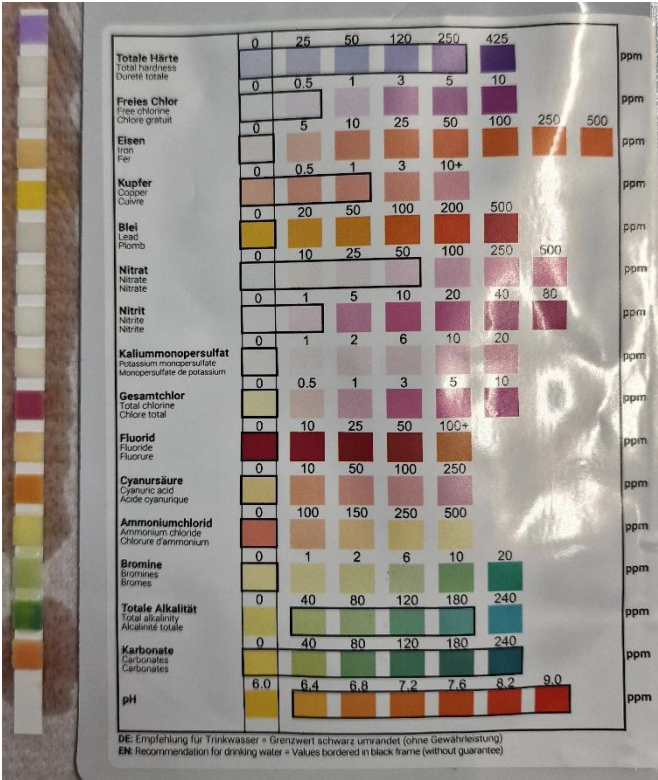
In Blantyre, two different water sources are available at the Innovation hub.

Zurich

Test: 01.07.2025

pH meter not functional → pH test strips used instead

Water sampling time: 15:09

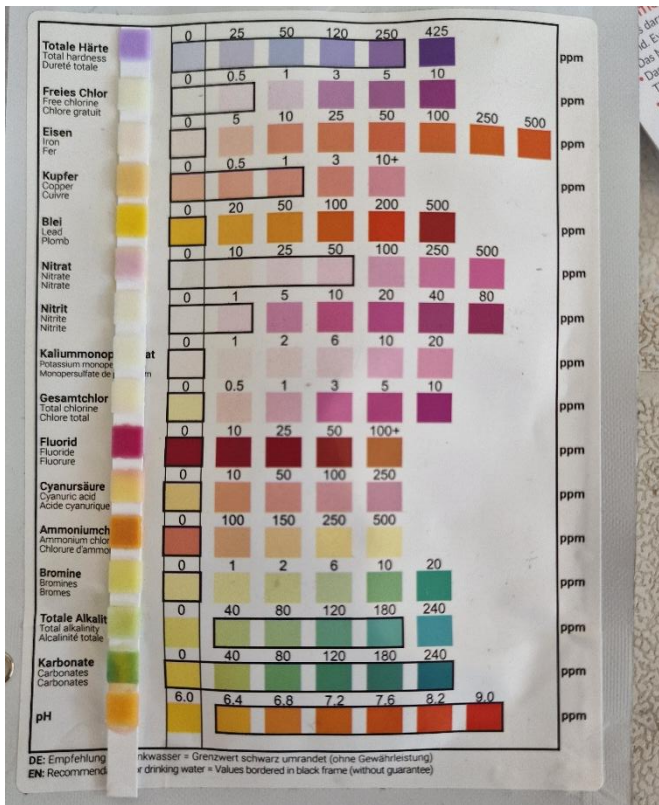
Test	Result	Observation
pH []	6 (repeated test with new pH meter on July 4, 2025: pH = 7.0 to 7.6 continuously growing)	
Water hardness [°dH]	10	10 drops
16 in 1 test [ppm]	Total hardness: 250 – 425 Free chlorine: 0 Iron: 0 Copper: 0 Lead: 0 Nitrate: 0 – 10 Nitrite: 0 Potassium monopersulfate: 0 – 1 Total chlorine: 0 Fluoride: 0 Cyanuric acid: 0 Ammonium chloride: 0 Bromines: 1 Total alkalinity: 40 Carbonates: 120 pH: 6.8	

Blantyre

Test 1: 23.09.2025 – Innovation Hub outside reservoir

pH meter not available → pH test strips used instead

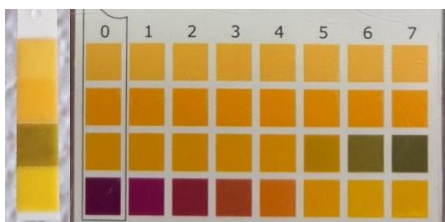
Water sampling time: 13:07 (test conducted 25.09.2025)

pH []	5 – 6	
Water hardness [°dH]	7	7 drops
16 in 1 test [ppm]	Total hardness: 250 Free chlorine: 0 Iron: 0 Copper: 0 Lead: 0 Nitrate: 50 – 100 Nitrite: 0 Potassium monopersulfate: 0 Total chlorine: 0 Fluoride: 0 Cyanuric acid: 0 Ammonium chloride: 0 – 100 Bromines: 1 Total alkalinity: 40 Carbonates: 40 – 80 pH: 6.4	

Test 2: 24.09.2025 – Innovation Hub inside tab

pH meter not available → pH test strips used instead

Water sampling time: 11:30 (test conducted 25.09.2025)

pH []	5 – 6	
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Water hardness [°dH]	6	6 drops
16 in 1 test [ppm]	Total hardness: 250 Free chlorine: 0 Iron: 0 Copper: 0 Lead: 0 Nitrate: 500 Nitrite: 10 Potassium monopersulfate: 0 Total chlorine: 0 Fluoride: 0 Cyanuric acid: 0 Ammonium chloride: 0 – 100 Bromines: 1 Total alkalinity: 40 Carbonates: 40 – 80 pH: 6.8	

Evaluation

The pH value is fairly similar between all measurements.

The hardness measurements returned higher values in Zurich than for the two measurements in Blantyre. As described in the explanations provided with the 16 in 1 test set (see Figure 1), a higher lime concentration reduces the solubility of soap in water. This could necessitate a larger amount of detergent in Zurich compared to Blantyre.

With the 16 in 1 test, much higher-than-normal nitrite and nitrate concentrations were detected in the water from the inside tap at Waste Adviser's Innovation Hub, while the nitrate value for the outside reservoir were only slightly above the recommended values for drinking water in Germany. This can be contributed to the high usage of fertilizer in Malawi's agriculture. The test in Zurich recorded nitrite and nitrate values well within these recommendations.

All other values were similar within the different water samples and mostly in a recommended range.

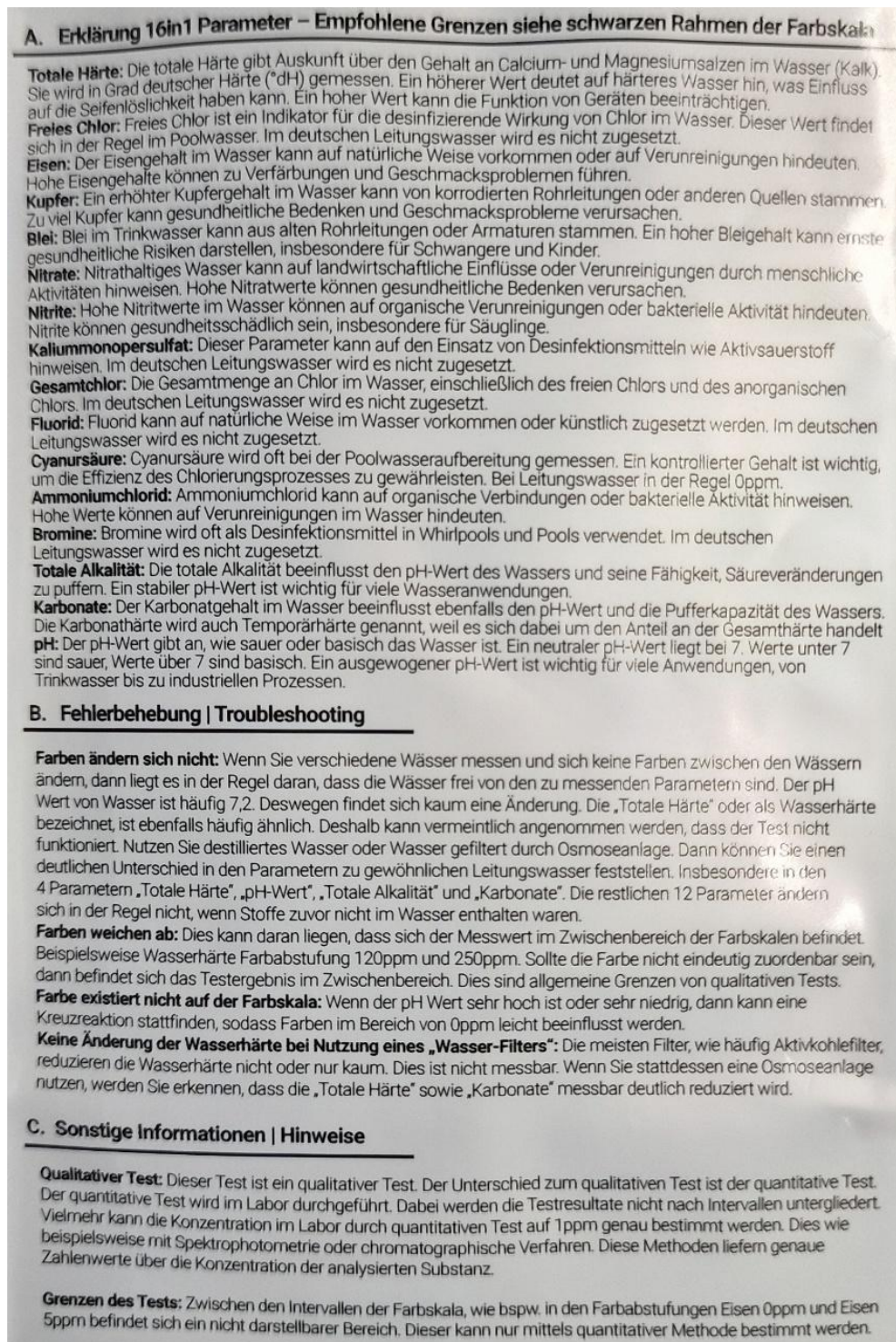


Figure 1: Explanations, troubleshooting, and other information enclosed with the 16 in 1 water quality tester.

Bibliography

Wasserhärte messen. (n.d.). *Wasserhärte Schweiz | Bögli AG Wasseraufbereitung*. Retrieved June 27, 2025, from <https://wasserhaerteschweiz.ch/messen/>